MILITARY SPECIFICATION

GAGES, PRESSURE, DIAL INDICATING

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design and all performance requirements for the procurement of direct indicating pressure gages used on aircraft and missiles.

1.2 Classification - The gages shall be of the following MS designations with the case size and dial scale as shown for the applicable MS part number specified.

MS Designation	Case Size In Inches	Dial Markings*	Dial Scale	MS Designation	Case Size In Inches	Dial Markings*	Dial Scale
MS17856-1	3/4	6	0-500	MS17857-9	1-1/2	10	0-2000
MS17856-2	3/4	8	0-1000	MS17857-10	1-1/2	11	0-3000
MS17856-3	3/4	10	0-2000	MS17857-11	1-1/2	12	0-4000
MS17856-4	3/4	11	0-3000	MS17857-12	1-1/2	13	0-5000
MS17856-5	3/4	12	0-4000	MS17857-13	2	1	0-50
MS17856-6	3/4	13	0-5000	MS17857-14	2	2	0-100
				MS17857-15	2	3	0-200
				MS17857-16	2	4	0-300
				MS17857-17	2	5	0-400
MS17857-1	1	6	0-500	MS17857-18	2	6	0-500
MS17857-2	1	8	0-1000	MS17857-19	2	7	0-600
MS17857-3	1	10	0-2000	MS17857-20	2	8	0-1000
MS17857-4	1	11	0-3000	MS17857-21	2	9	0-1500
MS17857-5	1	12	0-4000	MS17857-22	2	10	0-2000
MS17857-6	1	13	0-5000	MS17857-23	2	11	0-3000
MS17857-7	1-1/2	6	0-500	MS17857-24	2	12	0-4000
MS17857-8	1-1/2	8	0-1000	MS17857-25	2	13	0-5000

* See Figure 1.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein:

FSC 6685

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SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and Other Uses
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum selected series; General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-1	Standard for Laboratory Atmospheric Conditions for Testing
FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of the U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS17856	Gage, Pressure, Dial Indicating, Stainless Steel Case
MS17857	Gage, Pressure, Dial Indicating, Non-Ferrous Case

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MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard form of
MS33586	Metals, Definition of Dissimilar
MS33656	Fitting End, Standard Dimension for Flared Tube Connection and Gasket Seal
MS33639	Case, Instrument, Clamp Mounted, Aircraft

Air Force-Navy Aeronautical

AND 10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread

2.2 Availability of Documents -

(1) When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The gages furnished under this specification shall be products, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of Government Documents - Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.2.2 Standard Parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical Materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

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3.3.2 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the instrument may be subject when in storage or during normal service life.

3.3.2.1 Dissimilar Metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing.

3.3.3 Non-magnetic Materials - Non-magnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

3.3.4 Fungus-Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed case, fungicidal treatment will not be necessary.

3.3.5 Corrosive Fumes - Materials used in the interior construction of the instrument shall not produce corrosive or deleterious fumes.

3.4 Design and Construction - The gage shall be designed to meet the operational requirements of this specification. The instrument shall be so constructed that no parts will work loose in service and will withstand the normal strains of jars, vibrations and such other conditions as are incident to shipping, storage, installation and service without failure.

3.4.1 Maintenance - The construction shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures.

3.4.2 Case - The outline dimensions of the gage cases shall conform to MS33639 for the one inch, one and one half inch and two inch sizes. The dimension designated as L on MS33639 shall not exceed two inches for each gage. The cases of the gages MS17856 shall be made of series 300 stainless steel, shall be uniform in texture with a smooth surface and shall be rain and dust tight. The cases of the gages MS17857 shall be made of non-ferrous, low density metal, uniform in texture with a smooth surface and shall be rain and dust tight. The cases of the gages MS17857 shall be finished with a lusterless black material of Color No. 37038 of FED-STD-595. The finishing material shall be of a quality sufficient to ensure durability of the markings in service use.

3.4.3 Bosses - The tubing connection bosses for the MS17856 gages shall be in accordance with MS33656-4, style E. The bosses for the MS17857 gages shall be in accordance with AND 10050-4 with 0.688 inch wrench flats.

3.4.4 Pressure Relief Device - Each gage shall be provided with a device to relieve the internal case pressure in the event of Bourdon Tube failure. The device shall relieve the internal pressure at one half the value necessary to blow out the cover glass and shall permit sufficient flow so that internal case pressure will not exceed this value when subjected to instantaneous pneumatic pressures of twice the full scale reading of the gage.

3.4.5 Weight - The gages shall be so constructed that when completely assembled the weight shall not exceed 0.35 pound.

3.5 Dial -

3.5.1 Dial Plate - The dial plate shall be made of aluminum or aluminum alloy. The plate shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used to attach the dial plate, they shall be so located that they will not interfere with any dial marking or prevent locating any dial marking in the proper location.

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3.5.2 Visibility of Dial - The pointer, numerals, at least 1/16th inch of the shortest graduations and all other specified markings on the dial shall be visible from any point within the frustrum of a cone whose sides make an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.5.3 Cover Glass - The quality of the cover glass shall be in accordance with Specification DD-G-451, Type II, Quality AA. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicrons region.

3.5.4 Cover Glass Mounting - The distance between the front surface of the case and the outside surface of the cover glass shall not exceed .01 inch. The distance between the dial plate and the inside surface of the cover glass shall be the minimum practicable and shall not exceed .125 inch. The cover glass shall be properly sealed.

3.5.5 Dial Markings - The dial markings of each gage shall conform to Figure I, as listed above under Classification for the applicable MS Part number. The dimensions of the dial markings shall be in accordance with Table I. The style and proportions of the letters and numerals placed on the dial shall conform to MS33558. The background of each dial shall befinished with lusterless black material of Color No. 37038 of FED-STD-595. All markings on the dial shall be finished with lusterless white material of Color No. 37875 of FED-STD-595. The dial marking materials shall be of a quality sufficient to ensure durability of the markings in service use.

3.5.5.1 Numerals - Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial arc. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall not be less than 1/64th inch.

3.5.6 Pointer - The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. The pointer shall be firmly attached to the mechanism and shall be readily adjustable.

3.6 Performance - The gage shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.7 Physical Requirements -

3.7.1 Protective Treatment - When materials are used in the construction of an instrument that are subject to corrosion in atmospheric or environmental conditions likely to occur during normal service life, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.7.2 Magnesium Alloy Parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.7.3 Aluminum Alloy Parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625 except that the dial, if of aluminum alloy, small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with Specification MIL-C-5541.

3.7.4 Iron and Steel Parts - Where practicable, iron and steel parts shall be nickel, chromium or cadmium plated in accordance with Specifications QQ-N-290, QQ-C-320 and QQ-P-416, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium coated.

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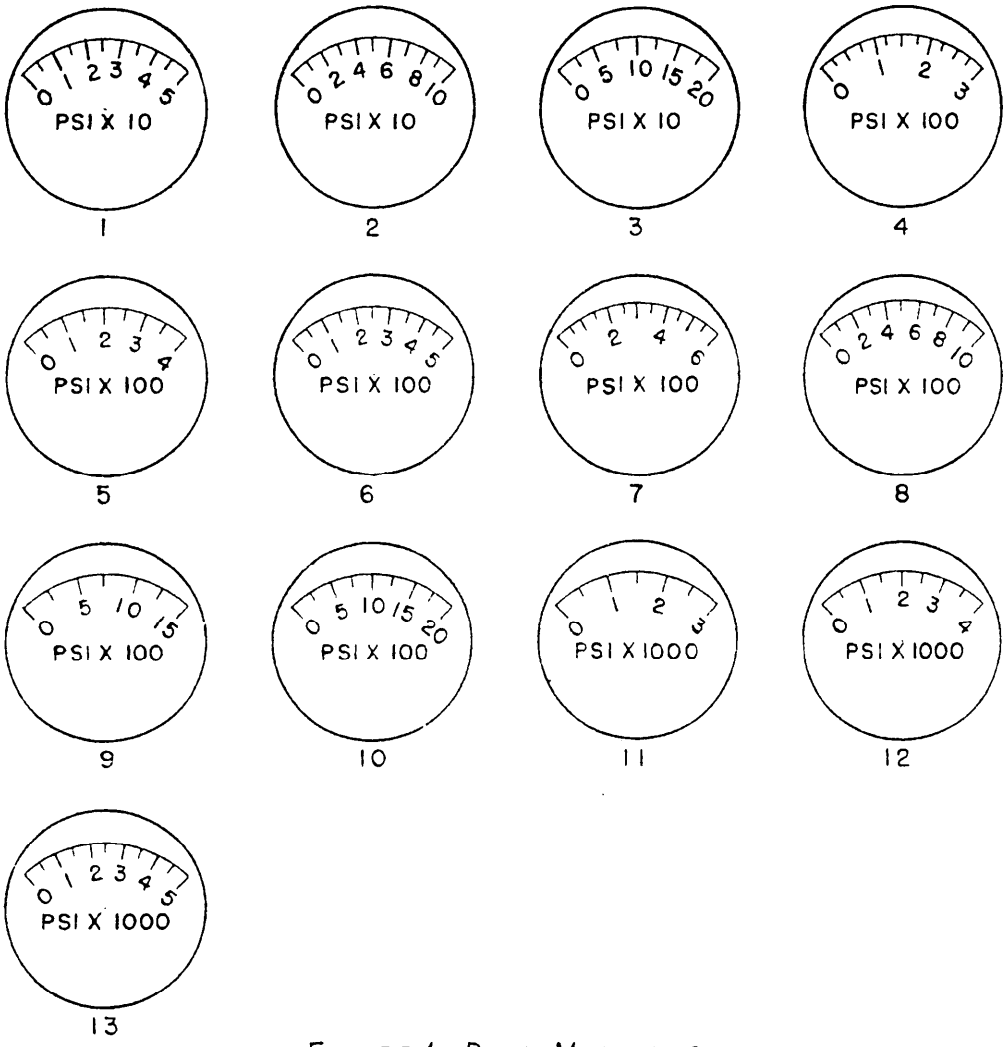


FIGURE 1. DIAL MARKINGS

TABLE 1. DIMENSIONS OF DIAL MARKINGS

	GAGE SIZE IN INCHES							
	3/4 DIA.		1 DIA.		1 1/2 DIA.		2 DIA.	
SCALE ARC	WIDTH ± .005	HEIGHT ± .010	WIDTH ± .005	HEIGHT ± .010	WIDTH ± .005	HEIGHT ± .010	WIDTH ± .005	HEIGHT ± .010
MAJOR GRADUATIONS	.015	.062	.015	.093	.015	.125	.015	.125
INTERMEDIATE GRADUATIONS	.010	.043	.010	.062	.010	.100	.010	.100
MINOR GRADUATIONS	.010	.031	.010	.043	.010	.075	.010	.075
NUMERALS	N/A	1/32	N/A	1/16	N/A	7/64	N/A	.125

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3.7.5 Screw Threads - Screw threads 0.060 inch in diameter or larger shall be in accordance with Specification MIL-S-7742.

3.7.5.1 Assembly Screws - Assembly screws and bolts shall be tight. The word "tight" means that the screws, bolts and/or nuts cannot be appreciably tightened further, without damage or injury to the threads.

3.7.6 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

3.7.7 Lubrication - The lubricating oil used shall be in accordance with Specification MIL-L-6085 and shall be sufficient for at least 1000 hours of operation.

3.7.8 Dimensions - Dimensions and tolerances not specified, shall be as close as is consistant with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation of performance of the instrument, they shall be held or limited accordingly.

3.7.9 Workmanship - The instrument, including all parts and accessories, shall be so constructed and finished that it will be free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, plating, riveting, machine-screw assemblies and freedom of parts from burrs and sharp edges.

3.7.10 Cleaning - The instrument shall be thoroughly cleaned of loose, spattered or excess solder, rosin flash that may crumble, metal chips and other foreign matter, after final assembly.

3.8 Identification and Installation -

3.8.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service. The nameplate shall be legibly marked in accordance with MIL-STD-130.

3.8.2 Manufacturer's Part Number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.8.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.8.4 Use of Military Designations - Military designations shall not be applied to a product, except for preproduction test samples, nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification.

3.8.5 Installation Instructions - The contractor shall furnish with each gage one copy of the Installation Instructions with illustrations and diagrams. Prior to printing, two copies shall be furnished to the procuring activity for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches.

3.8.6 Envelope - An envelope furnished by the contractor, containing installation instructions shall be packaged with each instrument and shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS
INSTALLATION INSTRUCTIONS

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tubes. The designators are limited to [illegible]

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The Inspection and Testing of the gages shall be classified as follows:

(a) Preproduction Tests - Preproduction tests are those tests performed after award of contract on sample instruments that are representative in design, performance and configuration of the item which will be produced on the manufacturer's production line.

(b) Acceptance Tests - Acceptance tests are those tests performed on instruments manufactured and submitted for acceptance under contract.

4.3 Preproduction Tests - The preproduction tests of the gages shall consist of all the tests of this specification specified under 4.6 Test Methods.

4.3.1 Preproduction Test Instructions - The preproduction test samples shall consist of three gages, of each manufacturer's part number to be furnished in accordance with the contract or order. The samples submitted shall not have been previously tested except for the Individual Tests. The manufacturer shall submit certified test results showing conformance with all the requirements and tests of this specification on samples other than the ones submitted for preproduction testing. The samples and certified test report shall be submitted to the testing activity designated in the contract or order.

4.3.2 Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below.

Sample for Preproduction Testing
Gage, Pressure, Dial Indicating
Submitted by (MANUFACTURER'S NAME, DATE) for Preproduction
Testing in accordance with Specification MIL-G-23337
under contract or order number _____.
MS _____ (Add proper dash number)
Manufacturer's Part Number

4.4 Acceptance Tests - The acceptance tests of the instruments shall consist of the Individual Tests and the Sampling Plans of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by the specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.4.1 Individual Tests - The individual tests shall consist of those listed below, conducted in the order given, on each gage.

Inspection
Scale Error at Room Temperature
Friction
Leakage
Overpressure

4.4.2 Sampling Plans and Tests -

4.4.2.1 Sampling Plan A - Samples shall consist of three gages selected at random by the Inspector from each lot of 100 or less which have passed the Individual Tests. A lot shall consist of identical instruments with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same conditions and submitted at substantially the same time. Instruments which have been subjected to the Sampling Plan A test shall not be delivered on contract until they have been reconditioned and resubmitted to all Individual Tests.

4.4.2.1.1 Sampling Plan A Tests - The sampling plan A tests shall consist of those listed below, conducted in the order given, on each sample.

Vibration
Low Temperature Exposure
High Temperature Exposure
Temperature Shock
Temperature-High Altitude

4.4.2.2 Sampling Plan B - Unless otherwise specified, the samples shall consist of three gages selected at random by the Inspector from the first 15 instruments produced on contract within ten days after manufacture. Samples which have been subjected to the Sampling Plan B tests shall not be delivered on contract until they have been reconditioned and resubmitted to all the Individual Tests. The samples shall be forwarded to the laboratory specified herein (see 6.2) and shall be plainly identified by securely attached, durable tags marked with the following information:

_____ (Name of item)
Submitted by (manufacturer's name, date) for production acceptance testing in accordance with Contract/Order No. _____
Manufacturer's Part Number

4.4.2.2.1 Sampling Plan B Testing - The sampling plan B tests shall consist of those listed below, conducted in the order given, on each sample.

Altitude
Shock
Endurance

4.4.3 Rejection and Retest - The instructions on reworking, reconditioning, refurbishing or resubmitting rejected items shall be as specified in the General Provisions of the Contract or Order.

4.5 Test Conditions -

4.5.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of FED-STD-1.

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4.5.2 Test Readings - Except where otherwise specified, the gage shall be vibrated before a test reading is taken. An electrical vibrator set at 30, 60, or 120 cps with a maximum amplitude of 0.002 inch when attached to the test item, shall be used.

4.5.3 Vibration Stand - Whenever a vibration stand is specified, it shall be a device which will vibrate at any desired frequency between 5 and 500 cps and shall subject the instrument to such vibration that a point on the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.5.4 Level Position - Unless otherwise specified, the instrument shall be tested in its normal operating position, with the dial vertical.

TABLE II

Gauge Pressure	Applied Test Press Beginning With 0 and Ending With Full Scale Range	Scale Error At Room Temperature After All Tests Except Endurance and Vibration	Scale Error At High Temp. Low Temp. and After Endurance and Vibration	Friction Error After All Tests
0-50	Every 10 PSI	±2 PSI	±4 PSI	±1 PSI
0-100	Every 20 PSI	±2 PSI	±4 PSI	±1 PSI
0-200	Every 50 PSI	±4 PSI	±6 PSI	±2 PSI
0-300	Every 50 PSI	±6 PSI	±10 PSI	±2 PSI
0-400	Every 50 PSI	±8 PSI	±10 PSI	±3 PSI
0-500	Every 100 PSI	±10 PSI	±15 PSI	±3 PSI
0-600	Every 100 PSI	±12 PSI	±15 PSI	±4 PSI
0-1000	Every 200 PSI	±20 PSI	±30 PSI	±5 PSI
0-1500	Every 250 PSI	±30 PSI	±40 PSI	±10 PSI
0-2000	Every 250 PSI	±40 PSI	±50 PSI	±15 PSI
0-3000	Every 500 PSI	±60 PSI	±90 PSI	±15 PSI
0-4000	Every 500 PSI	±80 PSI	±110 PSI	±20 PSI
0-5000	Every 1000 PSI	±100 PSI	±150 PSI	±25 PSI

4.6 Test Methods -

4.6.1 Inspection Test - Each gage shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

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4.6.2 Scale Error at Room Temperature Test - The instrument shall be tested for scale error at the pressures listed in Table II. The test shall be made by subjecting the instrument to the pressures specified to produce these readings first with the pressures increasing, then with the pressure decreasing. With the pressures increasing, the pressure shall be brought up to, but shall not exceed, the pressure specified to give the desired reading; and with the pressure decreasing, the pressure shall be brought down to, but shall not fall below, the pressure specified to give the desired reading. The scale error shall not exceed the tolerances specified in Table II.

4.6.3 Friction Test - The instrument shall be tested for friction at each alternate test point on increasing readings indicated in Table II beginning with the second test point. The pressure shall be increased slowly to bring the pointer approximately to the desired reading, and then held constant while two readings are taken, the first before the gage is vibrated, the second after the gage is vibrated. This test may be combined with the test for scale error.

4.6.4 Leakage Test - Pressure of 80% of range shall be applied to the gage. The connection of tubing shall then be pinched off or otherwise completely sealed at a point within 2 inches of the pressure connection. During a period of 10 minutes, the pointer shall be observed for deviation from the original pressure setting. Any drop of pressure that can be attributed to leaks in the bourdon tube shall be cause for rejection.

4.6.5 Overpressure Test - The applied pressure during this test shall be 150% of the rated pressure for each type. After the testing procedure the instrument shall be subjected to and meet the requirements specified in the following tests.

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

(3) Leakage Test

4.6.6 Vibration Test - The gage shall be tested in accordance with procedure XII of Specification MIL-E-5272 except that the G level shall be 20 G. At the end of the test the instrument shall be subjected to and meet the requirements specified in the following tests.

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

(3) Leakage Test

4.6.7 Low Temperature Exposure Test - The gage shall be tested in accordance with Procedure II of Specification MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

4.6.8 High Temperature Exposure Test - The gage shall be tested in accordance with Procedure II of Specification MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

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4.6.9 Temperature Shock Test - The gage shall be tested in accordance with procedure I of Specification on MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests.

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

4.6.10 Low Temperature-High Altitude Test - The gage shall be tested in accordance with Procedure I of Specification MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

4.6.11 Altitude Test - The gage, properly connected, shall be tested in accordance with Procedure VI, Condition E of Specification MIL-E-5272 for a period of one hour. During the last 5 minutes of the one hour period and with the temperature and pressure maintained as specified the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (Except Tolerance ± 6%)

(2) Friction Test

4.6.12 Shock Test - The gage shall be tested in accordance with Procedure IV of Specification MIL-E-5272 except the shock force shall be 100 G.

4.6.13 Endurance Test - The gage shall be subjected to 100,000 applications of pressure in a range which produces an indication from 0 to 80 ± 5 percent of the range of the gage at a rate of 30 ± 10 cycles per minute. The application and release of pressure shall be as smooth as practicable, so as not to subject the instrument mechanism to excessive acceleration and show approximate sign wave shape. The gage shall be tested for scale errors not less than one hour after the pressure cycling, the errors shall not exceed the tolerances specified in Table II. Any pressure gage subjected to the endurance test shall be examined to determine if the moving parts have been worn to the extent that premature failure may occur during actual service conditions. Units subjected to this test shall not be serviceable and will not be accepted on contract. After the testing procedure, the gage shall be subjected to and shall meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

(3) Leakage Test

4.6.14 Humidity Test - The gage shall be tested in accordance with Procedure III of Specification MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

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4.6.15 Fungus Resistance Test - The gage shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the procuring activity, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled gage.

4.6.16 Salt Spray Test - The gage shall be tested in accordance with Procedure I of Specification MIL-E-5272. After the testing procedure, the instrument shall be subjected to and meet the requirements specified in the following tests:

(1) Scale Error at Room Temperature Test (See Table II)

(2) Friction Test

5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A - The gage shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method 1A5, omitting preservation compound.

5.1.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing -

5.2.1 Levels A and B - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulations in effect at the time of shipment.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Procautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The gages covered by this specification are intended for use on aircraft to indicate directly the pressure in aircraft hydraulic and pneumatic system or pneumatic accumulators of hydraulic systems.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, Number and Date of this Specification

(b) The quantity and MS part number of the gage desired.

(c) Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted.

(d) Levels of packaging (see 5.1) and packing (see 5.2) desired.

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6.3 Drawing Requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.5 Provisions for Preproduction Tests - The manufacture of instruments on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the gages from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.5.1 It shall be understood that the instrument supplied under a contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by the implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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